# Technical Supplement to

Exact Diffusion Design for Maximally Collective Stable Turing Patterns
in Binary-Complex Mass-Action Networks

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## Contents



## 1 Reaction matrices, flux cone, and semipositive conservation

Order species as $X_1, \dots, X_m, Z$ and reactions as $R_0, R_2, \dots, R_{m-2}, R_a, R_b, R_+, R_-$. For each indexed reaction, the source and target vectors have total molecularity at most two. The source matrix $Y_m$ and stoichiometric matrix $\Gamma_m$ are obtained by placing these source vectors and reaction differences in columns.

The balance equations $\Gamma_m v = 0$ read, successively, equality of every flux around the long circuit and equality of the two boundary-pair fluxes. Thus

$$\ker \Gamma_m = \operatorname{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}.$$

An $m \times m$ minor formed by deleting one circuit reaction and one member of the reversible pair is triangular after elementary row operations and has absolute determinant $2^{m-2}$. Therefore $\operatorname{rank} \Gamma_m = m$. The vector

$$c = (0, 4, \dots, 4, 2, 1)^T$$

is orthogonal to every reaction vector, so $\operatorname{im} \Gamma_m = c^{\perp}$. Notice that $c$ is semipositive rather than positive: $X_1$ is absent.

For flux $(a\mathbf{1}_m, b, b)$ and unit equilibrium, $A_m(a,b) = \Gamma_m \operatorname{diag}(a\mathbf{1}_m, b, b) Y_m^T$ has precisely the entries printed in the main text. At an arbitrary positive equilibrium, right multiplication by $H = \operatorname{diag}(1/x_i^*)$ gives $J = A_m(a,b)H$. Conversely the rate assignment $k_r = v_r/(x^*)^{y_r}$ realizes every $a, b, H > 0$.

## 2 All-spectrum principal-block classification

Use the directed convention $X_j \to X_i$ when the $(i,j)$ Jacobian entry is nonzero. The interior arrows are the one-way chain $X_2 \to X_3 \to \cdots \to X_{m-1}$. The boundary arrows are those incident to $X_1, X_m, Z$ and the returns $X_{m-1} \to X_1$, $X_m \to X_2$. The following case analysis is exhaustive for an induced set with fewer than $m$ vertices.

1. If a component contains a chain arrow and returns through $X_{m-1} \to X_1$, then the unique forward path from $X_1$ forces every vertex $X_1, \ldots, X_{m-1}$. This is the first complete long cycle.
2. If it contains a chain arrow and returns through $X_m \to X_2$, then the forward path forces every vertex $X_2, \ldots, X_m$. This is the second complete long cycle.
3. The alternative chain closure $X_1 \to X_2 \to \cdots \to X_m \to X_1$ uses $m$ vertices and is excluded. Adding any boundary vertex to either complete long cycle also uses at least $m$ vertices.
4. If neither long cycle is complete, every feedback path not involving a complete chain lies inside $\{X_1, X_m, Z\}$. All other retained chain vertices have no return path and therefore form negative singleton diagonal blocks.

At $b = 2a$, the coefficient $A_{m1}$ vanishes. This deletes the arrow $X_1 \to X_m$; edge deletion can split but cannot merge strongly connected components, so the same list remains exhaustive.

For the boundary triad, the Routh–Hurwitz difference expands as

$$\begin{aligned} c_1c_2 - c_3 = {} & 4a^3h_1^2h_m + 16a^3h_1h_m^2 + 4a^2bh_1^2h_m + 23a^2bh_1h_m^2 \\ & + 16a^2bh_1h_mh_Z + 4a^2bh_1^2h_Z + 64a^2bh_m^2h_Z \\ & + 7ab^2h_1^2h_m + 7ab^2h_1h_m^2 + 28ab^2h_1h_mh_Z \\ & + 4ab^2h_1^2h_Z + 20ab^2h_1h_mh_Z + 16ab^2h_m^2h_Z + 64ab^2h_mh_Z^2. \end{aligned}$$

Every monomial is strictly positive. The two long-cycle modulus inequalities are strict because $a+b > a$ and $4a+b > 4a$. Frobenius permutation therefore proves Hurwitz stability of every smaller principal block.

Sparse expansion on $\{X_1, \ldots, X_m\}$ gives

$$(-1)^m \det J_{\widehat{Z},\widehat{Z}} = -2a^{m-1}b \prod_{i=1}^{m} h_i.$$

The complete omission table is derived in Section S4 below.

## 3 The principal-minor diffusion-ray theorem in full detail

Let $a_I = (-1)^{|I|} \det J_{I,I}$ and $a_\varnothing = 1$. Assume $\det J = 0$, $a_I > 0$ for $|I| \le n-2$, and $\sum_{|I|=n-1} a_I > 0$. The last condition follows from a simple zero and stable complementary spectrum but is stated directly because it is all the proof uses. Multilinearity in the diagonal entries of $sD$ gives

$$\det(sD - J) = \sum_{I \subseteq [n]} a_I \prod_{j \notin I} sd_j.$$

Regrouping by $k = n - |I|$ yields

$$\det(sD - J) = \sum_{k=1}^{n} \beta_k(D)s^k, \qquad \beta_k(D) = \sum_{|I|=n-k} a_I \prod_{j \notin I} d_j.$$

For $k \ge 2$, every contributing set has order at most $n-2$, so the stated coefficient hypothesis gives $a_I > 0$ and hence $\beta_k > 0$.

The bracket $q_D(s) = \beta_1 + \beta_2 s + \cdots + \beta_n s^{n-1}$ is strictly increasing for $s > 0$. If $\beta_1 < 0$, it has exactly one positive zero and its derivative there is positive. Since $\det(sD - J) = \det(D)\det(sI - D^{-1}J)$, the zero is algebraically simple.

For the positive-real-eigenvalue band, differentiate

$$\chi_s(\lambda) = \sum_I a_I \prod_{j \notin I} (\lambda + sd_j).$$

Every order-at-most-$(n-2)$ contribution is nonnegative for $\lambda \geq 0$. The order-$(n-1)$ contribution to $\chi_s'$ is the constant $\sum_{|I|=n-1} a_I$. This is positive by hypothesis. Under the common spectral corollary it equals the coefficient of $s$ in $\det(sI - J)$, and equal damping sends the simple zero and all stable eigenvalues left. Thus $\chi_s' > 0$ on the nonnegative real axis. The sign of $\chi_s(0)$ then determines whether there is exactly one positive real eigenvalue.

The theorem says nothing about nonreal unstable eigenvalues when $s > s_*$. The all-mode certificates below are separate, profile-specific results.

# 4 Order-$(n-1)$ omission minors and diffusion design

Right multiplication by $H$ multiplies a principal determinant by the product of its retained $h_i$, so the combinatorial coefficients may be computed at $H = I$.

Omitting $Z$ leaves the complete $X$-chain block. Its sparse cycle-cover expansion gives

$$(-1)^m \det J_{\widehat{Z},\widehat{Z}} = -2a^{m-1} b \prod_{i=1}^{m} h_i.$$

For $2 \leq j \leq m-1$, omit $X_j$ and order the remaining vertices by the two feed-forward chain fragments, followed by $\{X_1, X_m, Z\}$. The resulting matrix is block triangular. The $m-3$ chain singletons contribute signed factors $ah_i$, and the boundary triad contributes

$$(-1)^3 \det J_{\{X_1,X_m,Z\},\{X_1,X_m,Z\}} = 16a^2 b\, h_1 h_m h_Z.$$

Therefore

$$(-1)^m \det J_{\widehat{X_j},\widehat{X_j}} = 16a^{m-1} b h_Z \prod_{\substack{1\leq i\leq m \\ i\neq j}} h_i.$$

Omitting $X_1$ leaves the restricted semipositive conservation vector in the left kernel. Omitting $X_m$ leaves the restriction of $H^{-1}(2,-2,\ldots,-2,0,1)^T$ in the right kernel, because the omitted coordinate is zero. Hence both determinants vanish.

Multiplication by the omitted diffusivities produces

$$\beta_1(D) = 2a^{m-1} b \left(\prod_{i=1}^{m} h_i\right)\left(8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j} - d_Z\right).$$

This yields the exact diffusion criterion and explains the coefficient eight as the exact ratio 16/2.

Define the stable realization domain

$$\mathfrak{S}_m = \{(a,b,H) : a,b > 0,\ H \succ 0,\ A_m(a,b)H|_{c^\perp} \text{ is Hurwitz}\}.$$

The complete region $\mathfrak{S}_m$ is not characterized here. For any $(a,b,H) \in \mathfrak{S}_m$, if $d_{\min} = \min_i d_i$, then

$$d_Z > 8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j} \geq d_{\min} T(H),$$

so $\chi_D > T(H)$. Setting all $X$ diffusivities to one and $d_Z = T(H) + \varepsilon$ proves the nonattained fixed-$H$ infimum. Since $h_Z/h_j \geq 1/\chi_H$, one obtains $\chi_D \chi_H > 8(m-2)$. The implication $T(H) > 1$ is necessary under homogeneous stability; its converse is not proved.

# 5 Improved unit-equilibrium critical profile

Put $K_i = 91m - 181 - i$. The right vector and diffusion profile are

$$r_1 = 1, \quad r_i = -\frac{K_i}{63(m-2)}, \quad r_m = -\frac{2}{9}, \quad r_Z = \frac{5}{14},$$

$$d_1 = \frac{23}{63}, \quad d_i = \frac{1}{K_i}, \quad d_m = \frac{1}{7}, \quad d_Z = \frac{16}{45}.$$

Direct substitution gives $(A_m - D_m)r = 0$. With $\ell_Z = 1$, the left vector is

$$\ell_1 = -\frac{266}{815}, \quad \ell_i = \frac{78260(m-2)}{163(91m-180-i)}, \quad \ell_m = \frac{18368}{7335}, \quad \ell_Z = 1.$$

Define

$$\mathcal{H}_m = \sum_{j=1}^{m-2} \frac{1}{91m-181-j}.$$

Then

$$\ell^T r = -\frac{7043400m - 13600927 - 7043400\mathcal{H}_m}{924210} < 0,$$

$$\ell^T D_m r = -\frac{2(1760850\mathcal{H}_m + 16559)}{462105} < 0.$$

The harmonic bound $\mathcal{H}_m \le (m-2)/(90m-179)$ makes the first sign an elementary shifted-polynomial inequality.

### Homogeneous and spatial determinants

The homogeneous determinant has

$$\det(\lambda I - A_m) = \lambda q_m(\lambda), \qquad \lambda q_m = (1+\lambda)^{m-3}P(\lambda) - R(\lambda),$$

with $P = \lambda^4 + 12\lambda^3 + 42\lambda^2 + 47\lambda + 16$ and $R = 5\lambda^2 + 33\lambda + 16$. For a mode with damping factor $t$,

$$\det(\lambda I - A_m + tD_m) = Q_m F - G,$$

where

$$Q_m = \prod_{i=2}^{m-1}\left(\lambda + 1 + \frac{t}{K_i}\right),$$

$$g_1 = \lambda + 2 + \frac{23}{63}t, \quad g_m = \lambda + 5 + \frac{1}{7}t, \quad g_Z = \lambda + 4 + \frac{16}{45}t,$$

$$F = g_1 g_m g_Z - 4g_1 - 4g_m + g_Z, \qquad G = g_Z(4g_1 + g_m) - 36.$$

At onset, $\prod_{i=2}^{m-1}(1 + K_i^{-1}) = 91/90$. The exact certificate tables appear in Section S8.

# 6 Conservation-compatible stable-mode corrections

The quadratic tensor is obtained reaction by reaction. Its components are

$$\begin{aligned}
(B(u,v))_1 &= -(u_1 v_{m-1} + u_{m-1} v_1) + 2u_Z v_Z - (u_1 v_m + u_m v_1),\\
(B(u,v))_2 &= -(u_1 v_2 + u_2 v_1) + 2u_m v_m,\\
(B(u,v))_i &= u_1 v_{i-1} + u_{i-1} v_1 - u_1 v_i - u_i v_1,\\
(B(u,v))_m &= 2(u_1 v_{m-1} + u_{m-1} v_1) - 4u_m v_m + 2u_Z v_Z - (u_1 v_m + u_m v_1),\\
(B(u,v))_Z &= -4u_Z v_Z + 2(u_1 v_m + u_m v_1).
\end{aligned}$$

Here the third line holds for $3 \le i \le m-1$. Reaction-wise conservation implies $c^T B(u,v) = 0$.

The zero mode solves

$$A_m w_0 = -\frac{1}{4}B(r,r), \qquad c^T w_0 = 0.$$

The exact solution is

$$(w_0)_1 = \frac{182448m - 373417}{31752(8m-17)},$$

$$(w_0)_2 = \frac{1008m^2 - 20459m + 37138}{31752(m-2)(8m-17)},$$

$$(w_0)_i = (w_0)_2 - \frac{i-2}{126(m-2)}, \quad (w_0)_m = -\frac{1}{81}, \quad (w_0)_Z = \frac{16861m - 34044}{7938(8m-17)}.$$

The second harmonic solves

$$(A_m - 4D_m)w_2 = -\frac{1}{4}B(r,r).$$

With

$$\sigma = \frac{1}{126(m-2)}, \qquad T_i = \frac{K_{i-3}K_{i-2}K_{i-1}K_i}{K_{-1}K_0K_1K_2},$$

the chain recurrence is

$$(w_2)_i = T_i\left((w_2)_2 + \frac{\sigma K_2}{3}\right) - \frac{\sigma K_i}{3}.$$

Its boundary denominator is

$$Q_m = 589180301m^3 - 3500015940m^2 + 6930529579m - 4574434500.$$

After $m = u + 3$, the coefficients are

$$589180301, \quad 1802606769, \quad 1838302066, \quad 624878904,$$

so it never vanishes for $m \geq 3$. The complete boundary values are provided in the machine-readable certificate and in the public verifier.

The cubic coefficient is

$$c_m = \frac{\ell^T\{B(r,w_0) + \frac{1}{2}B(r,w_2)\}}{\ell^T r}.$$

The numerator equals $R_m + C_m\mathcal{H}_m$, where $R_m > 0$ and $C_m < 0$. Substitution of $\mathcal{H}_m \leq (m-2)/(90m-179)$ reduces positivity to the degree-five polynomial

$$\begin{aligned} L_m = {} & 272994514782766788672 0m^5 - 277551324204741709999952m^4 \\ & + 112813395868533457497683m^3 - 229153280695458887386228m^2 \\ & + 232620996871721820873517m - 94412163900120968220300. \end{aligned}$$

After $m = u + 3$, its coefficients are

$$\begin{gathered} 27299451478276678867 20, 131940447969408473008 48, 254468701273335153030 59, \\ 244753213292073255099 11, 117364846083668751203 74, 224393377003330583848 8, \end{gathered}$$

all positive. Hence the cubic numerator is positive and, since $\ell^T r < 0$, $c_m < 0$.

# 7 Equilibrium-scaled stable family

Let $\nu = m - 2$ and $L \in [1/\sqrt{3\nu}, 90\nu/(90\nu+1)]$. The lower endpoint is the sufficient boundary of the present modulus certificate $\nu L^2 \geq 1/3$, not a proved intrinsic dynamical boundary. Set

$$h_1 = h_m = h_Z = 1, \qquad h_i = \frac{K_i}{LK_{i-1}}.$$

The physical equilibrium is $x^* = H^{-1}\mathbf{1}$ and the physical diffusion matrix is $D^{\text{phys}} = H\Delta$, where $\Delta$ is the improved unit profile. In normalized variables $z = Hx$,

$$z_t = H\{f_m(z) + \Delta z_{\xi\xi}\}.$$

The stationary bracket and critical right vector are unchanged. The dynamic left vector is $H^{-1}\ell$ and the physical conservation gauge uses $H^{-1}c$.

For interior chain factors,

$$g_i = \frac{K_{i-1}}{K_i}(1 + L\lambda) + \frac{t-1}{K_i}.$$

When $t \geq 1$ and $\Re\lambda \geq 0$,

$$|Q_m|^2 \geq \left(\frac{91}{90}\right)^2 |1 + L\lambda|^{2\nu} \geq \left(\frac{91}{90}\right)^2 \left(1 + 2\nu L\Re\lambda + \frac{1}{3}(\Im\lambda)^2\right).$$

The 34- and 84-term coefficient tables in Section S8 prove homogeneous and all-mode stability.

Let $\rho_m = (2, -2, \ldots, -2, 0, 1)^T$. The physical zero-mode gauge is

$$w_0(L) = w_0^{\mathrm{ref}} + \tau_m(L)\rho_m, \qquad \tau_m(L) = -\frac{(H^{-1}c)^T w_0^{\mathrm{ref}}}{(H^{-1}c)^T \rho_m}.$$

The second harmonic is unchanged. The cubic numerator satisfies

$$N_m(L) = N_m^{\mathrm{ref}} + \tau_m(L) S_m,$$

$$N_m^{\mathrm{ref}} > \frac{1}{100}, \qquad -\frac{1}{10} < S_m < 0, \qquad \tau_m(L) < \frac{1}{20},$$

so $N_m(L) > 1/200$. Also

$$\ell^T H^{-1} r = -\frac{485873}{924210} - \frac{11180}{1467} L(m-2) < 0.$$

Therefore the family is uniformly supercritical over the certified interval.

The contrasts are

$$\chi_D(L) = \frac{23}{63}(91\nu L), \qquad \chi_H(L) = \frac{91\nu - 1}{91\nu L},$$

with constant product $23(91\nu - 1)/63$. At $L = 1/\sqrt{3\nu}$, both have square-root growth.

# 8 Exact coefficient tables

Each row below records the exponent vector and exact coefficient of the corresponding polynomial. In the Pareto tables, the coefficient column is a polynomial in $A = 2\nu L \Re\lambda$ (or any positive lower coefficient replacing it). All listed coefficients are nonnegative; the strict first-order rows identify the unique equality case. The command `python verify_symbolic_certificates.py` regenerates and checks the tables from exact symbolic expressions.

### 35-term homogeneous certificate

| $\deg_x$ | $\deg_z$ | coefficient |
|---|---|---|
| 10 | 0 | 1 |
| 9 | 0 | 26 |
| 8 | 1 | 5 |
| 8 | 0 | 277 |
| 7 | 1 | 104 |
| 7 | 0 | 1582 |
| 6 | 2 | 10 |
| 6 | 1 | 892 |
| 6 | 0 | 5356 |
| 5 | 2 | 156 |
| 5 | 1 | 4034 |
| 5 | 0 | 11282 |
| 4 | 3 | 10 |
| 4 | 2 | 1014 |
| 4 | 1 | 10432 |
| 4 | 0 | 15116 |
| 3 | 3 | 104 |
| 3 | 2 | 3322 |
| 3 | 1 | 15628 |
| 3 | 0 | 12612 |
| 2 | 4 | 5 |
| 2 | 3 | 460 |
| 2 | 2 | 5804 |
| 2 | 1 | 13296 |
| 2 | 0 | 5568 |

| $\deg_x$ | $\deg_z$ | coefficient |
|---|---|---|
| 1 | 4 | 26 |
| 1 | 3 | 870 |
| 1 | 2 | 4858 |
| 1 | 1 | 5700 |
| 1 | 0 | 960 |
| 0 | 5 | 1 |
| 0 | 4 | 61 |
| 0 | 3 | 728 |
| 0 | 2 | 1508 |
| 0 | 1 | 192 |

**77-term improved-profile spatial certificate**

| $\deg_x$ | $\deg_z$ | $\deg_s$ | coefficient |
|---|---|---|---|
| 6 | 1 | 0 | 1 |
| 6 | 0 | 0 | $\frac{8281}{8100}$ |
| 5 | 1 | 1 | $\frac{544}{315}$ |
| 5 | 1 | 0 | $\frac{7474}{315}$ |
| 5 | 0 | 1 | $\frac{160888}{91125}$ |
| 5 | 0 | 0 | $\frac{4420871}{182250}$ |
| 4 | 2 | 0 | 3 |
| 4 | 1 | 2 | $\frac{40058}{33075}$ |
| 4 | 1 | 1 | $\frac{1147126}{33075}$ |
| 4 | 1 | 0 | $\frac{9681647}{44100}$ |
| 4 | 0 | 2 | $\frac{3384901}{2733750}$ |
| 4 | 0 | 1 | $\frac{96932147}{2733750}$ |
| 4 | 0 | 0 | $\frac{1073317517}{5467500}$ |
| 3 | 2 | 1 | $\frac{1088}{315}$ |
| 3 | 2 | 0 | $\frac{14948}{315}$ |
| 3 | 1 | 3 | $\frac{2744576}{6251175}$ |
| 3 | 1 | 2 | $\frac{41276762}{2083725}$ |
| 3 | 1 | 1 | $\frac{8186375738}{31255875}$ |
| 3 | 1 | 0 | $\frac{10434543841}{10418625}$ |
| 3 | 0 | 3 | $\frac{115958336}{258339375}$ |
| 3 | 0 | 2 | $\frac{3487886389}{172226250}$ |
| 3 | 0 | 1 | $\frac{39671949511}{172226250}$ |
| 3 | 0 | 0 | $\frac{315432259813}{516678750}$ |
| 2 | 3 | 0 | 3 |
| 2 | 2 | 2 | $\frac{147968}{99225}$ |
| 2 | 2 | 1 | $\frac{4065856}{99225}$ |
| 2 | 2 | 0 | $\frac{112938659}{396900}$ |
| 2 | 1 | 4 | $\frac{33947257}{393824025}$ |
| 2 | 1 | 3 | $\frac{2164641406}{393824025}$ |
| 2 | 1 | 2 | $\frac{1130666508023}{9845600625}$ |
| 2 | 1 | 1 | $\frac{9041228678666}{9845600625}$ |
| 2 | 1 | 0 | $\frac{5074377793279}{2187911250}$ |
| 2 | 0 | 4 | $\frac{5737086433}{65101522500}$ |
| 2 | 0 | 3 | $\frac{182912198807}{32550761250}$ |
| 2 | 0 | 2 | $\frac{2142193804481}{21700507500}$ |
| 2 | 0 | 1 | $\frac{17194597630733}{32550761250}$ |
| 2 | 0 | 0 | $\frac{34399269232129}{65101522500}$ |
| 1 | 3 | 1 | $\frac{544}{315}$ |
| 1 | 3 | 0 | $\frac{7474}{315}$ |
| 1 | 2 | 3 | $\frac{1817216}{6251175}$ |
| 1 | 2 | 2 | $\frac{24359162}{2083725}$ |

| $\deg_x$ | $\deg_z$ | $\deg_s$ | coefficient |
|---|---|---|---|
| 1 | 2 | 1 | $\frac{5056135954}{31255875}$ |
| 1 | 2 | 0 | $\frac{47437544293}{62511750}$ |
| 1 | 1 | 5 | $\frac{3399584}{393824025}$ |
| 1 | 1 | 4 | $\frac{57943048}{78764805}$ |
| 1 | 1 | 3 | $\frac{1934522477168}{88610405625}$ |
| 1 | 1 | 2 | $\frac{16308922895827}{59073603750}$ |
| 1 | 1 | 1 | $\frac{85574440190593}{59073603750}$ |
| 1 | 1 | 0 | $\frac{62146993882837}{25317258750}$ |
| 1 | 0 | 5 | $\frac{143632424}{16275380625}$ |
| 1 | 0 | 4 | $\frac{2448093778}{3255076125}$ |
| 1 | 0 | 3 | $\frac{98527787884}{5425126875}$ |
| 1 | 0 | 2 | $\frac{475980699652}{3255076125}$ |
| 1 | 0 | 1 | $\frac{4488365319374}{16275380625}$ |
| 1 | 0 | 0 | $\frac{1086412028}{10333575}$ |
| 0 | 4 | 0 | 1 |
| 0 | 3 | 2 | $\frac{27794}{99225}$ |
| 0 | 3 | 1 | $\frac{624478}{99225}$ |
| 0 | 3 | 0 | $\frac{1747307}{26460}$ |
| 0 | 2 | 4 | $\frac{1744613}{78764805}$ |
| 0 | 2 | 3 | $\frac{437610814}{393824025}$ |
| 0 | 2 | 2 | $\frac{452220491743}{19691201250}$ |
| 0 | 2 | 1 | $\frac{4035916860091}{19691201250}$ |
| 0 | 2 | 0 | $\frac{10861695961357}{13127467500}$ |
| 0 | 1 | 6 | $\frac{135424}{393824025}$ |
| 0 | 1 | 5 | $\frac{2964608}{78764805}$ |
| 0 | 1 | 4 | $\frac{962538972893}{637994920500}$ |
| 0 | 1 | 3 | $\frac{43240103983367}{1594987301250}$ |
| 0 | 1 | 2 | $\frac{238300444324673}{1063324867500}$ |
| 0 | 1 | 1 | $\frac{170939874507539}{227855328750}$ |
| 0 | 1 | 0 | $\frac{32638690751329}{65101522500}$ |
| 0 | 0 | 6 | $\frac{5721664}{16275380625}$ |
| 0 | 0 | 5 | $\frac{125254688}{3255076125}$ |
| 0 | 0 | 4 | $\frac{6479423564}{5425126875}$ |
| 0 | 0 | 3 | $\frac{41256502232}{3255076125}$ |
| 0 | 0 | 2 | $\frac{513831327844}{16275380625}$ |
| 0 | 0 | 1 | $\frac{148103696}{10333575}$ |

**34-term equilibrium-scaled homogeneous certificate**

| $\deg_x$ | $\deg_z$ | coefficient |
|---|---|---|
| 9 | 0 | $A$ |
| 8 | 1 | $\frac{1}{3}$ |
| 8 | 0 | $1+24A$ |
| 7 | 1 | $8+4A$ |
| 7 | 0 | $24+228A$ |
| 6 | 2 | $\frac{4}{3}$ |
| 6 | 1 | $80+72A$ |
| 6 | 0 | $228+1102A$ |
| 5 | 2 | $24+6A$ |
| 5 | 1 | $\frac{1318}{3}+516A$ |
| 5 | 0 | $1102+2924A$ |
| 4 | 3 | 2 |
| 4 | 2 | $178+72A$ |
| 4 | 1 | $\frac{4472}{3}+1828A$ |
| 4 | 0 | $2899+4332A$ |

| $\deg_x$ | $\deg_z$ | coefficient |
|---|---|---|
| 3 | 3 | $24 + 4A$ |
| 3 | 2 | $\frac{2044}{3} + 348A$ |
| 3 | 1 | $3272 + 3336A$ |
| 3 | 0 | $4002 + 3553A$ |
| 2 | 4 | $\frac{4}{3}$ |
| 2 | 3 | $120 + 24A$ |
| 2 | 2 | $1460 + 726A$ |
| 2 | 1 | $\frac{13411}{3} + 2796A$ |
| 2 | 0 | $2304 + 1504A$ |
| 1 | 4 | $8 + A$ |
| 1 | 3 | $266 + 60A$ |
| 1 | 2 | $1658 + 668A$ |
| 1 | 1 | $\frac{8902}{3} + 865A$ |
| 1 | 0 | $448 + 256A$ |
| 0 | 5 | $\frac{1}{3}$ |
| 0 | 4 | $21$ |
| 0 | 3 | $\frac{848}{3}$ |
| 0 | 2 | $\frac{2794}{3}$ |
| 0 | 1 | $\frac{64}{3}$ |

**84-term equilibrium-scaled spatial certificate**

| $\deg_x$ | $\deg_z$ | $\deg_s$ | coefficient |
|---|---|---|---|
| 7 | 0 | 0 | $\frac{8281}{8100}A$ |
| 6 | 1 | 0 | $\frac{8281}{24300}$ |
| 6 | 0 | 1 | $\frac{160888}{91125}A$ |
| 6 | 0 | 0 | $\frac{8281}{8100} + \frac{4420871}{182250}A$ |
| 5 | 1 | 1 | $\frac{160888}{273375}$ |
| 5 | 1 | 0 | $\frac{4420871}{546750} + \frac{8281}{2700}A$ |
| 5 | 0 | 2 | $\frac{3384901}{2733750}A$ |
| 5 | 0 | 1 | $\frac{160888}{91125} + \frac{96932147}{2733750}A$ |
| 5 | 0 | 0 | $\frac{4420871}{182250} + \frac{1210005017}{5467500}A$ |
| 4 | 2 | 0 | $\frac{8281}{8100}$ |
| 4 | 1 | 2 | $\frac{3384901}{8201250}$ |
| 4 | 1 | 1 | $\frac{96932147}{8201250} + \frac{321776}{91125}A$ |
| 4 | 1 | 0 | $\frac{315078023}{4100625} + \frac{4420871}{91125}A$ |
| 4 | 0 | 3 | $\frac{115958336}{258339375}A$ |
| 4 | 0 | 2 | $\frac{3384901}{2733750} + \frac{3487886389}{172226250}A$ |
| 4 | 0 | 1 | $\frac{96932147}{2733750} + \frac{45494837011}{172226250}A$ |
| 4 | 0 | 0 | $\frac{1073317517}{5467500} + \frac{503404909813}{516678750}A$ |
| 3 | 2 | 1 | $\frac{321776}{273375}$ |
| 3 | 2 | 0 | $\frac{4420871}{273375} + \frac{8281}{2700}A$ |
| 3 | 1 | 3 | $\frac{115958336}{775018125}$ |
| 3 | 1 | 2 | $\frac{3487886389}{516678750} + \frac{6251648}{4100625}A$ |
| 3 | 1 | 1 | $\frac{47319306931}{516678750} + \frac{171782416}{4100625}A$ |
| 3 | 1 | 0 | $\frac{578603925523}{1550036250} + \frac{2360113561}{8201250}A$ |
| 3 | 0 | 4 | $\frac{5737086433}{65101522500}A$ |
| 3 | 0 | 3 | $\frac{115958336}{258339375} + \frac{182912198807}{32550761250}A$ |
| 3 | 0 | 2 | $\frac{3487886389}{172226250} + \frac{2513945531981}{21700507500}A$ |
| 3 | 0 | 1 | $\frac{39671949511}{172226250} + \frac{29165273350733}{32550761250}A$ |
| 3 | 0 | 0 | $\frac{315432259813}{516678750} + \frac{138537167092129}{65101522500}A$ |
| 2 | 3 | 0 | $\frac{8281}{8100}$ |
| 2 | 2 | 2 | $\frac{6251648}{12301875}$ |
| 2 | 2 | 1 | $\frac{171782416}{12301875} + \frac{160888}{91125}A$ |
| 2 | 2 | 0 | $\frac{4871148347}{49207500} + \frac{4420871}{182250}A$ |

| $\deg_x$ | $\deg_z$ | $\deg_s$ | coefficient |
|---|---|---|---|
| 2 | 1 | 4 | $\frac{5737086433}{195304567500}$ |
| 2 | 1 | 3 | $\frac{182912198807}{97652283750} + \frac{76777376}{258339375}A$ |
| 2 | 1 | 2 | $\frac{2613196695629}{65101522500} + \frac{2058349189}{172226250}A$ |
| 2 | 1 | 1 | $\frac{33256099805357}{97652283750} + \frac{28172026051}{172226250}A$ |
| 2 | 1 | 0 | $\frac{184975683058783}{195304567500} + \frac{388034017813}{516678750}A$ |
| 2 | 0 | 5 | $\frac{143632424}{16275380625}A$ |
| 2 | 0 | 4 | $\frac{5737086433}{65101522500} + \frac{2448093778}{3255076125}A$ |
| 2 | 0 | 3 | $\frac{182912198807}{32550761250} + \frac{119438423884}{5425126875}A$ |
| 2 | 0 | 2 | $\frac{2142193804481}{21700507500} + \frac{878963880652}{3255076125}A$ |
| 2 | 0 | 1 | $\frac{17194597630733}{32550761250} + \frac{21944279796374}{16275380625}A$ |
| 2 | 0 | 0 | $\frac{34399269232129}{65101522500} + \frac{21842305628}{10333575}A$ |
| 1 | 3 | 1 | $\frac{160888}{273375}$ |
| 1 | 3 | 0 | $\frac{4420871}{546750} + \frac{8281}{8100}A$ |
| 1 | 2 | 3 | $\frac{76777376}{775018125}$ |
| 1 | 2 | 2 | $\frac{2058349189}{516678750} + \frac{2348593}{8201250}A$ |
| 1 | 2 | 1 | $\frac{29084261011}{516678750} + \frac{52768391}{8201250}A$ |
| 1 | 2 | 0 | $\frac{212816762834}{775018125} + \frac{1090212071}{16402500}A$ |
| 1 | 1 | 5 | $\frac{143632424}{48826141875}$ |
| 1 | 1 | 4 | $\frac{2448093778}{9765228375} + \frac{294839597}{13020304500}A$ |
| 1 | 1 | 3 | $\frac{124275398572}{16275380625} + \frac{36978113783}{32550761250}A$ |
| 1 | 1 | 2 | $\frac{9956722796683}{97652283750} + \frac{503148512477}{21700507500}A$ |
| 1 | 1 | 1 | $\frac{11312104230233}{19530456750} + \frac{6606581721077}{32550761250}A$ |
| 1 | 1 | 0 | $\frac{241757054977}{221433750} + \frac{52308962411329}{65101522500}A$ |
| 1 | 0 | 6 | $\frac{5721664}{16275380625}A$ |
| 1 | 0 | 5 | $\frac{143632424}{16275380625} + \frac{125254688}{3255076125}A$ |
| 1 | 0 | 4 | $\frac{2448093778}{3255076125} + \frac{8242156364}{5425126875}A$ |
| 1 | 0 | 3 | $\frac{98527787884}{5425126875} + \frac{86436565592}{3255076125}A$ |
| 1 | 0 | 2 | $\frac{475980699652}{3255076125} + \frac{3438519744244}{16275380625}A$ |
| 1 | 0 | 1 | $\frac{4488365319374}{16275380625} + \frac{7094103536}{10333575}A$ |
| 1 | 0 | 0 | $\frac{1086412028}{10333575} + \frac{4999696}{6561}A$ |
| 0 | 4 | 0 | $\frac{8281}{24300}$ |
| 0 | 3 | 2 | $\frac{2348593}{24603750}$ |
| 0 | 3 | 1 | $\frac{52768391}{24603750}$ |
| 0 | 3 | 0 | $\frac{570259573}{24603750}$ |
| 0 | 2 | 4 | $\frac{294839597}{39060913500}$ |
| 0 | 2 | 3 | $\frac{36978113783}{97652283750}$ |
| 0 | 2 | 2 | $\frac{521791643711}{65101522500}$ |
| 0 | 2 | 1 | $\frac{3617447476357}{48826141875}$ |
| 0 | 2 | 0 | $\frac{30203751676613}{97652283750}$ |
| 0 | 1 | 6 | $\frac{5721664}{48826141875}$ |
| 0 | 1 | 5 | $\frac{125254688}{9765228375}$ |
| 0 | 1 | 4 | $\frac{11480941147}{21700507500}$ |
| 0 | 1 | 3 | $\frac{975299997269}{97652283750}$ |
| 0 | 1 | 2 | $\frac{17163191001769}{195304567500}$ |
| 0 | 1 | 1 | $\frac{10010796373877}{32550761250}$ |
| 0 | 1 | 0 | $\frac{650029271329}{65101522500}$ |
| 0 | 0 | 6 | $\frac{5721664}{16275380625}$ |
| 0 | 0 | 5 | $\frac{125254688}{3255076125}$ |
| 0 | 0 | 4 | $\frac{6479423564}{5425126875}$ |
| 0 | 0 | 3 | $\frac{41256502232}{3255076125}$ |
| 0 | 0 | 2 | $\frac{513831327844}{16275380625}$ |
| 0 | 0 | 1 | $\frac{148103696}{10333575}$ |

### Shifted-polynomial and cubic-comparison certificates

**Unit-profile boundary denominator.** After $m = z + 3$, the coefficients of the cubic denominator are

$$\begin{gathered}589180301, \quad 1802606769, \quad 1838302066, \\ 624878904\end{gathered}$$

**Unit-profile critical-denominator bound.** After $m = z + 3$, the auxiliary quadratic has coefficients

$$633906000, \quad 1311540570, \quad 678120443$$

**Unit-profile cubic numerator lower bound.** After $m = z + 3$, the decisive degree-five polynomial has coefficients

$$\begin{gathered}272994514782766788672 0, \quad 1319404479694084730084 8, \quad 2544687012733351530305 9, \\ 2447532132920732550991 1, \quad 1173648460836687512037 4, \quad 2243933770033305838488\end{gathered}$$

**Equilibrium-scaled coefficient sign.** After $m = z + 3$, the polynomial whose sign gives $S_m < 0$ has coefficients

$$\begin{gathered}652054120726848, \quad 2672677467393709, \quad 4108255612276161, \\ 2806739366867294, \quad 719107361052288\end{gathered}$$

**Reference cubic margin.** After $\nu = z + 1$, the numerator proving $N_m^{\mathrm{ref}} > 1/100$ has coefficients

$$\begin{gathered}3790502986637265684840, \quad 1797829840271772813771 1, \quad 3395506017705733448357 3, \\ 3189984359505146437929 2, \quad 1489518898634836803572 8, \quad 276261020755504372005 6\end{gathered}$$

**Gauge upper-bound tail.** For $\nu \geq 2$, the polynomial upper bound used in $\tau_m(L) < 1/20$ has coefficients

$$\begin{gathered}-189709065/2, \quad -507201030, \quad -935658, \\ 58481\end{gathered}$$

**Remaining scalar comparisons.** The certificate also checks exactly

$$\frac{1760850}{91} - 10253 > 0, \qquad \frac{4}{462105}\left(\frac{1760850}{90} - 10253\right) < \frac{1}{10},$$

which give $-1/10 < S_m < 0$, and combines $N_m^{\mathrm{ref}} > 1/100$, $S_m > -1/10$, and $\tau_m(L) < 1/20$ to obtain $N_m(L) > 1/200$.

# 9 Near-threshold control example

For the affine path

$$p = \varepsilon, \quad u = 1 + (2 - t)\varepsilon + \theta\varepsilon^2, \quad v = t\varepsilon - \theta\varepsilon^2,$$
$$q = \frac{1}{2} - \left(\frac{1}{2} + t\right)\varepsilon + \left(\theta - \frac{M}{2}\right)\varepsilon^2,$$

the critical vector tends to one half of the homogeneous kernel vector. The diffusion excess over the exact linear threshold is

$$d_Z - 8\sum_{j=2}^{m-1} d_j = 4\left(M + 6t + 3t^2 - \frac{t^2}{m-2}\right)\varepsilon^2 + O(\varepsilon^3).$$

For $(t, \theta, M) = (2/9, 1/2, 1)$ and $m = 3$, exact elimination yields

$$c_3(\varepsilon) = \frac{6}{1379} + \frac{421985}{11409846}\varepsilon + O(\varepsilon^2),$$

and rational remainder bounds prove $c_3(\varepsilon) > 0$ on $0 < \varepsilon \leq 10^{-3}$. This is a rigorous subcritical control example, not a universal nonlinear lower bound.

## 10 Semilinear stability and robustness

On the real fixed-integrated-mass $L^2$ space, the positive diagonal Neumann diffusion operator has domain $H^2_N$, is sectorial, and generates an analytic semigroup. Its half-order fractional domain is $H^1$. In one space dimension, $H^1$ is a Banach algebra and embeds into $C^0$, so the quadratic mass-action Nemytskii map is smooth from $H^1$ to $L^2$. The patterned branch lies in $H^2_N$. The fixed-mass restriction removes the homogeneous conservation zero, and Neumann boundary conditions on an interval leave no continuous translation-neutral mode. Strict spectral negativity therefore implies local exponential asymptotic stability in $H^1$.

For robustness, use one scalar diffusion multiplier as the implicit variable. The critical eigenvalue derivative is nonzero, so the implicit function theorem continues the crossing under small perturbations of rates, equilibrium coordinates, diffusion ratios, and interval length. Low-mode gaps persist by analytic perturbation. A positive diffusion lower bound and a reaction-Jacobian upper bound give one neighborhood controlling all high modes. Smooth dependence of the Lyapunov–Schmidt coefficients preserves the negative cubic sign and the branch spectral gap. This is a retuned codimension-one statement and is not a claim that every nearby point is itself at bifurcation.

## 11 Numerical protocol and independent verification

Cosine–Galerkin simulations use exact modal convolution for the quadratic reaction terms and an adaptive stiff integrator. Initial data are small perturbations of $\mathbf{1}$ in the critical direction. The first-mode amplitude is measured by the adjoint projection. Spatial and temporal refinement tables, open-format profile data, exact parameters, and all solver tolerances are included in the data archive. No numerical output is used in a proof.

The release contains independent programs for the reaction family, flux cone, all-spectrum block theorem, omission minors, principal-minor diffusion-ray theorem, diffusion criterion, contrast bounds, improved profile, Pareto family, mode certificates, harmonic corrections, cubic sign, and branch stability. Mutation tests change signs, endpoints, the factor eight, Fourier factors, and certificate coefficients and must be rejected.